2026 Queensland AI in Healthcare Symposium

# Reducing Cognitive Load in Clinical Decision-Making with GenAI-Powered Report Analysis

*– a prototype and study proposal*

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## BACKGROUND

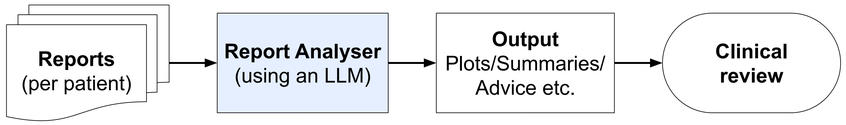
Clinicians managing complex patients must manually navigate multiple longitudinal test reports. It is a cognitively demanding process that makes it difficult to identify trends and synthesise findings at the point of care. The burden of integrating these findings can lead to missed insights and suboptimal clinical decisions.

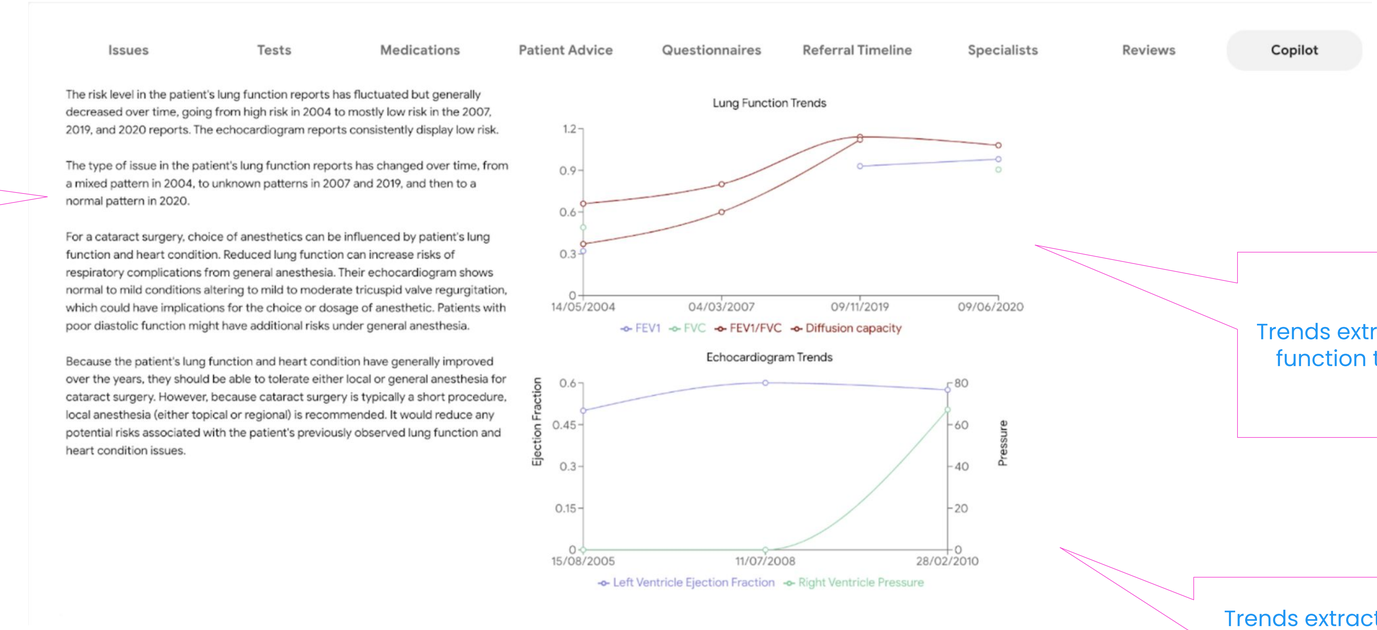
## AIMS

To develop a system that automatically extracts, summarises, and visualises trends from longitudinal patient test reports (structured and unstructured data), and surfaces personalised best-practice clinical recommendations.

## METHODS

The system uses a model-agnostic LLM as an analyser to iteratively process patients' test reports, extract data points for key clinical metrics, plot trends and create a summary. The prompt specified instructions to parse reports, expected metrics and guardrails to avoid hallucinations. Best-practice recommendations were codified and dynamically presented to the user, personalised for the patient, through a rule-based look-up mechanism. The system was tested on respiratory function tests and echocardiograms.

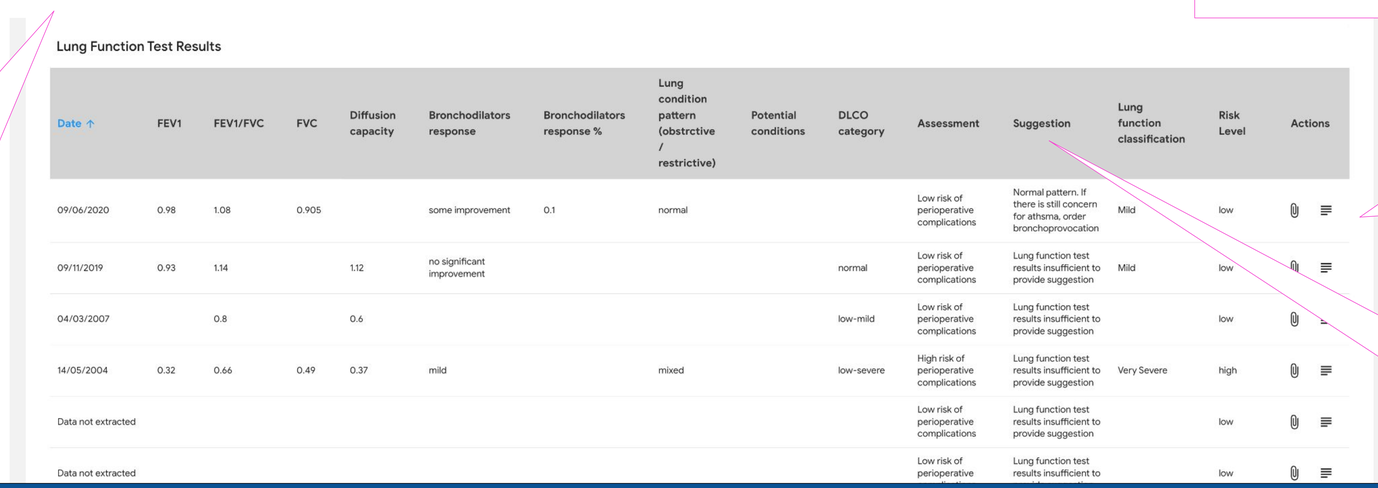



Automated summary

Trends extracted from lung function tests over time

Trends extracted from echos over time



Lung Function Test Results

| Date ↑ | FEV1 | FEV1/FVC | FVC | Diffusion capacity | Bronchodilators response | Bronchodilators response % | Lung condition pattern (obstrctive / restrictive) | Potential conditions | DLCO category | Assessment | Suggestion | Lung function classification | Risk Level | Actions |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 09/06/2020 | 0.98 | 1.08 | 0.905 | | some improvement | 0.1 | normal | | | Low risk of perioperative complications | Normal pattern. If there is still concern for athsma, order bronchoprovocation | Mild | low | |
| 09/11/2019 | 0.93 | 1.14 | | 1.12 | no significant improvement | | | | normal | Low risk of perioperative complications | Lung function test results insufficient to provide suggestion | Mild | low | |
| 04/03/2007 | | 0.8 | | 0.6 | | | | | low-mild | Low risk of perioperative complications | Lung function test results insufficient to provide suggestion | | low | |
| 14/05/2004 | 0.32 | 0.66 | 0.49 | 0.37 | mild | | mixed | | low-severe | High risk of perioperative complications | Lung function test results insufficient to provide suggestion | Very Severe | high | |
| Data not extracted | | | | | | | | | | Low risk of perioperative complications | Lung function test results insufficient to provide suggestion | | low | |
| Data not extracted | | | | | | | | | | Low risk of perioperative | Lung function test results insufficient to | | low | |

Tabular format of numerical AND **non-numerical** trends extracted from lung function tests.

Includes synthesis:
- Assessment
- Suggestions
- Classification
- Risk Level
- Actions

Users can download original report

Advice based on best-practice

## RESULTS

The system analysed reports, extracted key information, presented trends over time in plot and table formats, provided text-based summaries and gave best-practice clinical recommendations. Results were reviewed by an experienced anaesthetist and found to be accurate.

## CONCLUSIONS

This proof-of-concept demonstrates that a GenAI-based system can accurately synthesise longitudinal patient reports, visualise results and surface personalised best-practice recommendations, significantly reducing clinician cognitive load. A prospective clinical trial is warranted.

## FUTURE WORK

- RAG-based dynamic best-practice guideline lookup
- Clinician chat interface for interactive report discussion
- Prospective clinical trial evaluation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